Data Analytics (DA)

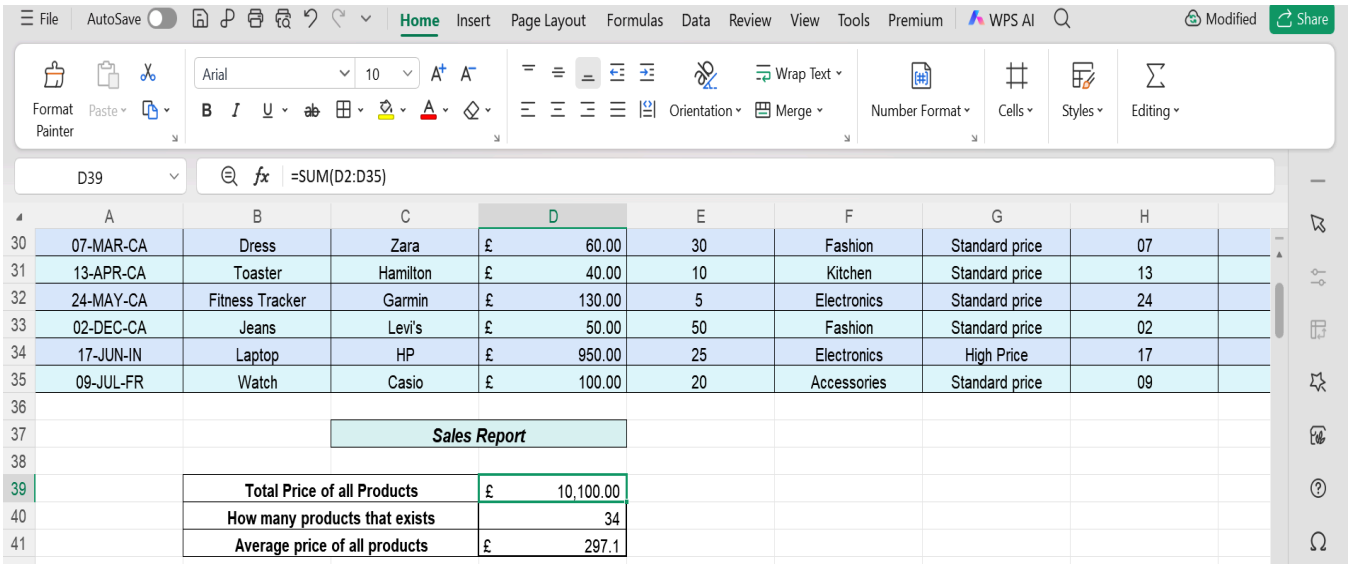
Module -1 - Excel

Excel Assignment 1 – Data Exploration

Basic Data Exploration

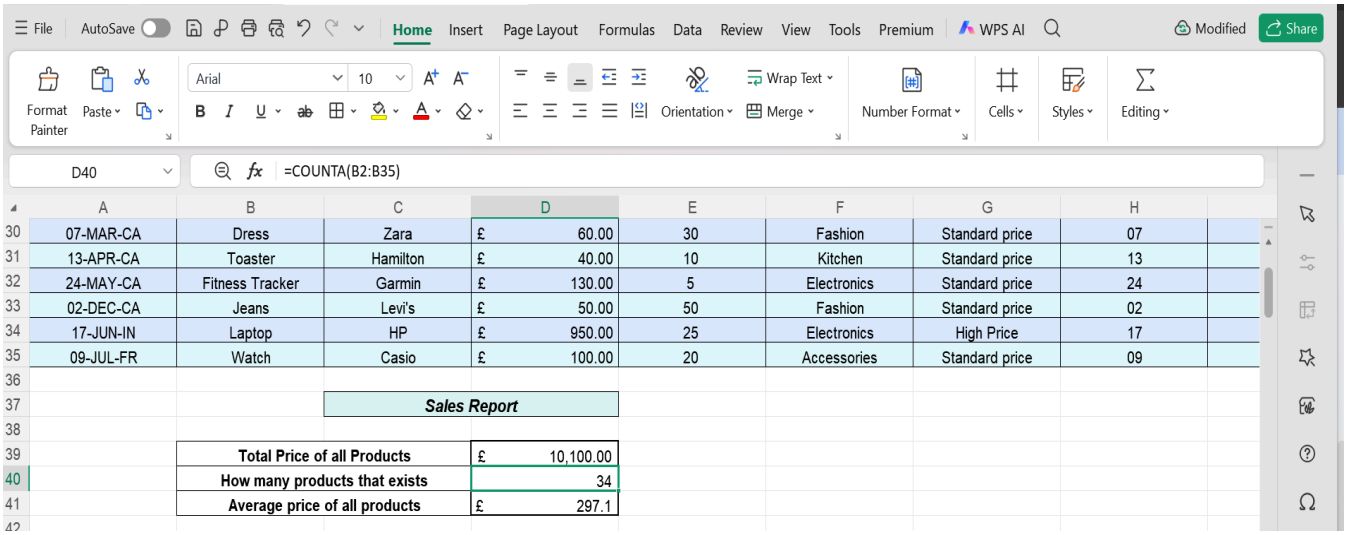
Step 1: Sum, Count, and Average

Total price of all products: (SUM).



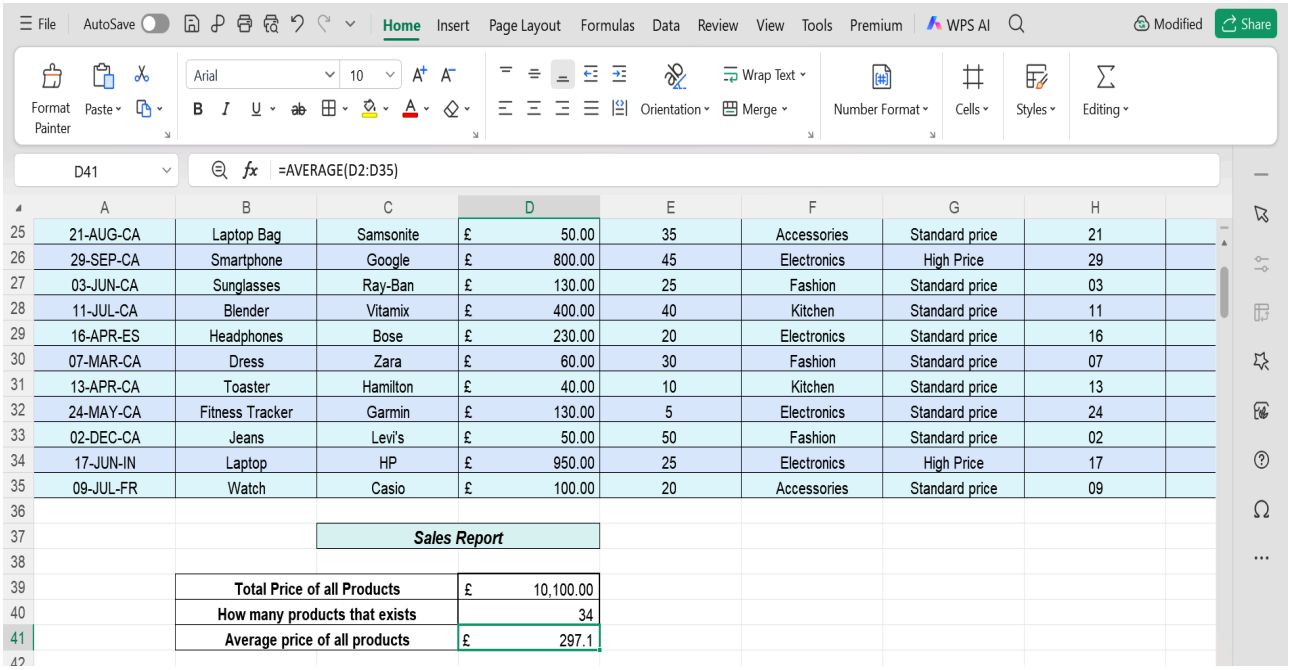
	A	B	C	D	E	F	G	H
30	07-MAR-CA	Dress	Zara	£ 60.00	30	Fashion	Standard price	07
31	13-APR-CA	Toaster	Hamilton	£ 40.00	10	Kitchen	Standard price	13
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	Electronics	Standard price	24
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	Fashion	Standard price	02
34	17-JUN-IN	Laptop	HP	£ 950.00	25	Electronics	High Price	17
35	09-JUL-FR	Watch	Casio	£ 100.00	20	Accessories	Standard price	09
36								
37			***Sales Report***					
38								
39		**Total Price of all Products**		£ 10,100.00				
40		**How many products that exists**		34				
41		**Average price of all products**		£ 297.1				

How many products exist in the dataset: (COUNTA).



	A	B	C	D	E	F	G	H
30	07-MAR-CA	Dress	Zara	£ 60.00	30	Fashion	Standard price	07
31	13-APR-CA	Toaster	Hamilton	£ 40.00	10	Kitchen	Standard price	13
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	Electronics	Standard price	24
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	Fashion	Standard price	02
34	17-JUN-IN	Laptop	HP	£ 950.00	25	Electronics	High Price	17
35	09-JUL-FR	Watch	Casio	£ 100.00	20	Accessories	Standard price	09
36								
37			***Sales Report***					
38								
39		**Total Price of all Products**		£ 10,100.00				
40		**How many products that exists**		34				
41		**Average price of all products**		£ 297.1				

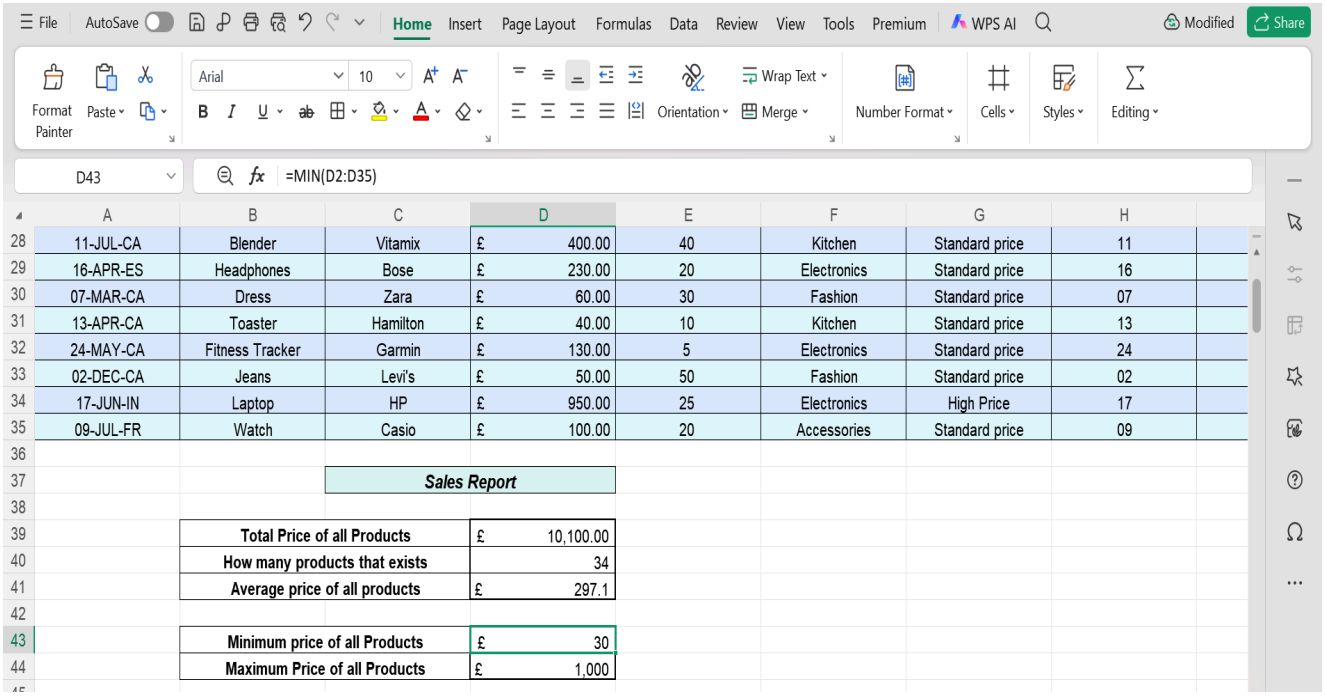
Average price of the products: (AVERAGE).



	A	B	C	D	E	F	G	H
25	21-AUG-CA	Laptop Bag	Samsonite	£ 50.00	35	Accessories	Standard price	21
26	29-SEP-CA	Smartphone	Google	£ 800.00	45	Electronics	High Price	29
27	03-JUN-CA	Sunglasses	Ray-Ban	£ 130.00	25	Fashion	Standard price	03
28	11-JUL-CA	Blender	Vitamix	£ 400.00	40	Kitchen	Standard price	11
29	16-APR-ES	Headphones	Bose	£ 230.00	20	Electronics	Standard price	16
30	07-MAR-CA	Dress	Zara	£ 60.00	30	Fashion	Standard price	07
31	13-APR-CA	Toaster	Hamilton	£ 40.00	10	Kitchen	Standard price	13
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	Electronics	Standard price	24
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	Fashion	Standard price	02
34	17-JUN-IN	Laptop	HP	£ 950.00	25	Electronics	High Price	17
35	09-JUL-FR	Watch	Casio	£ 100.00	20	Accessories	Standard price	09
36								
37			***Sales Report***					
38								
39		**Total Price of all Products**		£ 10,100.00				
40		**How many products that exists**		34				
41		**Average price of all products**		£ 297.1				

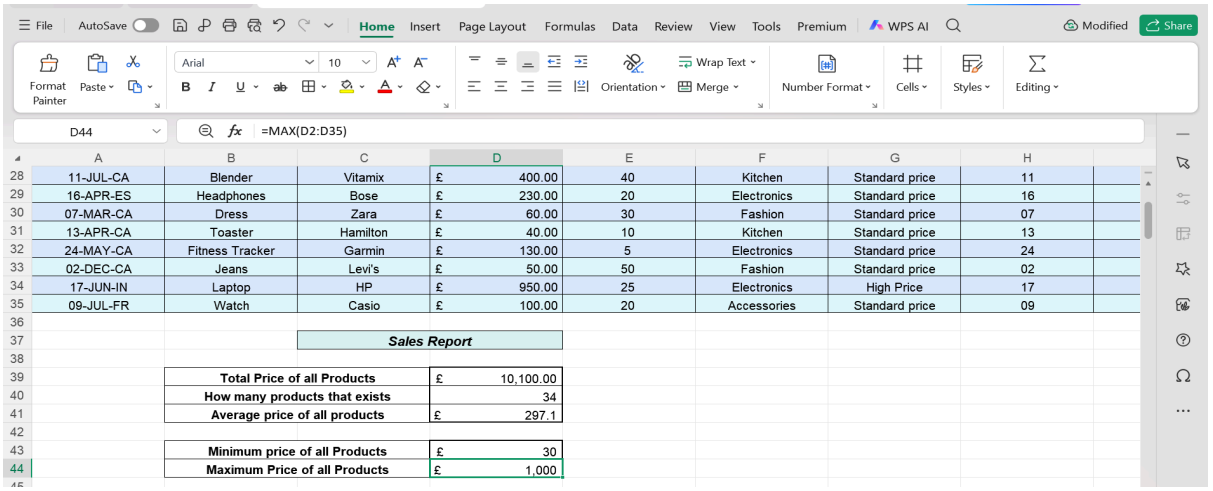
Step 2: Minimum and Maximum Values

Minimum price: (MIN).



	A	B	C	D	E	F	G	H
28	11-JUL-CA	Blender	Vitamix	£ 400.00	40	Kitchen	Standard price	11
29	16-APR-ES	Headphones	Bose	£ 230.00	20	Electronics	Standard price	16
30	07-MAR-CA	Dress	Zara	£ 60.00	30	Fashion	Standard price	07
31	13-APR-CA	Toaster	Hamilton	£ 40.00	10	Kitchen	Standard price	13
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	Electronics	Standard price	24
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	Fashion	Standard price	02
34	17-JUN-IN	Laptop	HP	£ 950.00	25	Electronics	High Price	17
35	09-JUL-FR	Watch	Casio	£ 100.00	20	Accessories	Standard price	09
36								
37			***Sales Report***					
38								
39		**Total Price of all Products**		£ 10,100.00				
40		**How many products that exists**		34				
41		**Average price of all products**		£ 297.1				
42								
43		**Minimum price of all Products**		£ 30				
44		**Maximum Price of all Products**		£ 1,000				

Maximum price: (MAX).

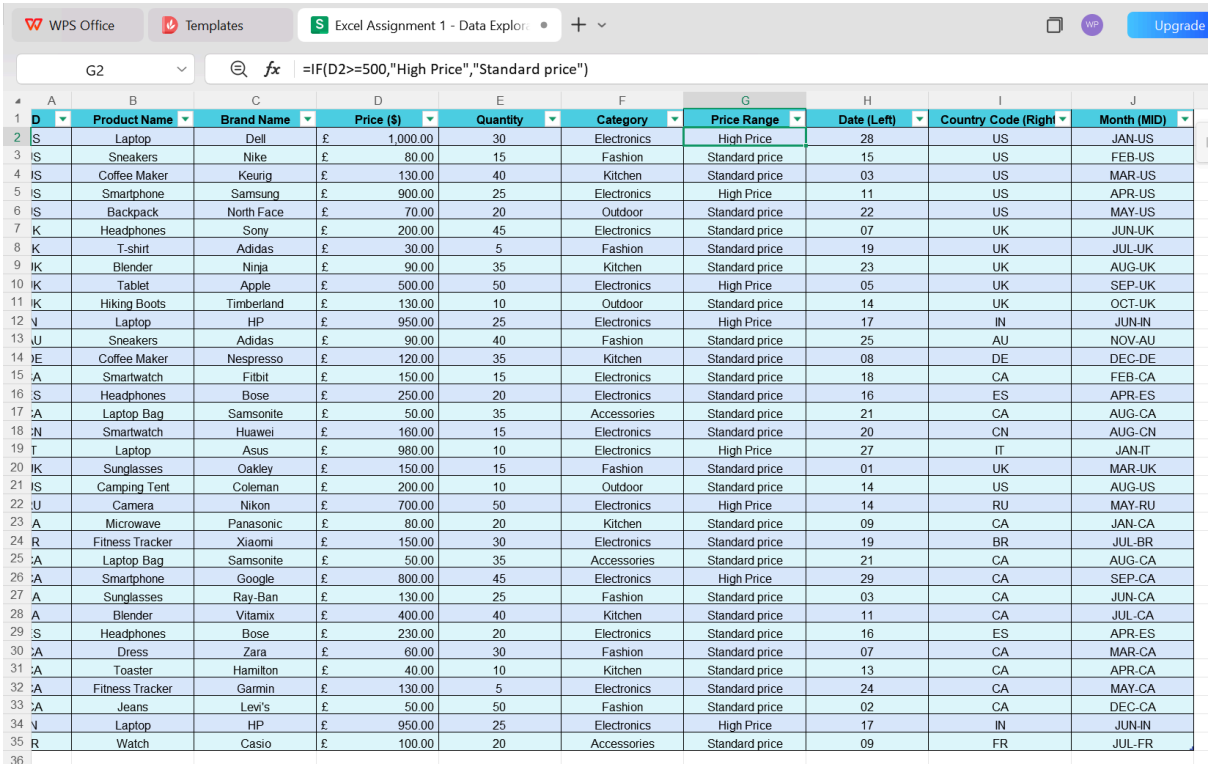


	A	B	C	D	E	F	G	H
28	11-JUL-CA	Blender	Vitamix	£ 400.00	40	Kitchen	Standard price	11
29	16-APR-ES	Headphones	Bose	£ 230.00	20	Electronics	Standard price	16
30	07-MAR-CA	Dress	Zara	£ 60.00	30	Fashion	Standard price	07
31	13-APR-CA	Toaster	Hamilton	£ 40.00	10	Kitchen	Standard price	13
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	Electronics	Standard price	24
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	Fashion	Standard price	02
34	17-JUN-IN	Laptop	HP	£ 950.00	25	Electronics	High Price	17
35	09-JUL-FR	Watch	Casio	£ 100.00	20	Accessories	Standard price	09
36								
37			***Sales Report***					
38								
39		**Total Price of all Products**		£ 10,100.00				
40		**How many products that exists**		34				
41		**Average price of all products**		£ 297.1				
42								
43		**Minimum price of all Products**		£ 30				
44		**Maximum Price of all Products**		£ 1,000				

Step 3: Logical Function – IF Condition

If Price ≥ $500 → classify as (High Price).

If Price < $500 → classify as (Standard Price).

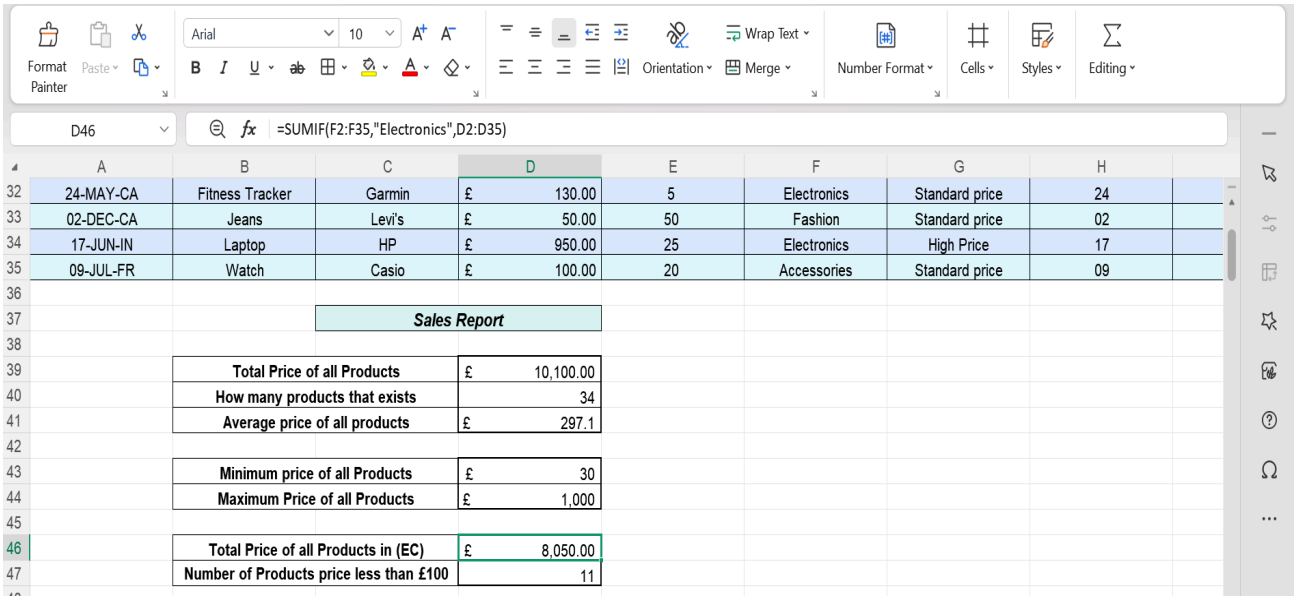


	A	B	C	D	E	F	G	H	I	J
1	D	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range	Date (Left)	Country Code (Righ	Month (MID)
2	S	Laptop	Dell	£ 1,000.00	30	Electronics	High Price	28	US	JAN-US
3	S	Sneakers	Nike	£ 80.00	15	Fashion	Standard price	15	US	FEB-US
4	S	Coffee Maker	Keurig	£ 130.00	40	Kitchen	Standard price	03	US	MAR-US
5	S	Smartphone	Samsung	£ 900.00	25	Electronics	High Price	11	US	APR-US
6	S	Backpack	North Face	£ 70.00	20	Outdoor	Standard price	22	US	MAY-US
7	K	Headphones	Sony	£ 200.00	45	Electronics	Standard price	07	UK	JUN-UK
8	K	T-shirt	Adidas	£ 30.00	5	Fashion	Standard price	19	UK	JUL-UK
9	K	Blender	Ninja	£ 90.00	35	Kitchen	Standard price	23	UK	AUG-UK
10	K	Tablet	Apple	£ 500.00	50	Electronics	High Price	05	UK	SEP-UK
11	K	Hiking Boots	Timberland	£ 130.00	10	Outdoor	Standard price	14	UK	OCT-UK
12	N	Laptop	HP	£ 950.00	25	Electronics	High Price	17	IN	JUN-IN
13	U	Sneakers	Adidas	£ 90.00	40	Fashion	Standard price	25	AU	NOV-AU
14	E	Coffee Maker	Nespresso	£ 120.00	35	Kitchen	Standard price	08	DE	DEC-DE
15	A	Smartwatch	Fitbit	£ 150.00	15	Electronics	Standard price	18	CA	FEB-CA
16	S	Headphones	Bose	£ 250.00	20	Electronics	Standard price	16	ES	APR-ES
17	A	Laptop Bag	Samsonite	£ 50.00	35	Accessories	Standard price	21	CA	AUG-CA
18	N	Smartwatch	Huawei	£ 160.00	15	Electronics	Standard price	20	CN	AUG-CN
19	T	Laptop	Asus	£ 980.00	10	Electronics	High Price	27	IT	JAN-IT
20	K	Sunglasses	Oakley	£ 150.00	15	Fashion	Standard price	01	UK	MAR-UK
21	S	Camping Tent	Coleman	£ 200.00	10	Outdoor	Standard price	14	US	AUG-US
22	U	Camera	Nikon	£ 700.00	50	Electronics	High Price	14	RU	MAY-RU
23	A	Microwave	Panasonic	£ 80.00	20	Kitchen	Standard price	09	CA	JAN-CA
24	R	Fitness Tracker	Xiaomi	£ 150.00	30	Electronics	Standard price	19	BR	JUL-BR
25	A	Laptop Bag	Samsonite	£ 50.00	35	Accessories	Standard price	21	CA	AUG-CA
26	A	Smartphone	Google	£ 800.00	45	Electronics	High Price	29	CA	SEP-CA
27	A	Sunglasses	Ray-Ban	£ 130.00	25	Fashion	Standard price	03	CA	JUN-CA
28	A	Blender	Vitamix	£ 400.00	40	Kitchen	Standard price	11	CA	JUL-CA
29	S	Headphones	Bose	£ 230.00	20	Electronics	Standard price	16	ES	APR-ES
30	A	Dress	Zara	£ 60.00	30	Fashion	Standard price	07	CA	MAR-CA
31	A	Toaster	Hamilton	£ 40.00	10	Kitchen	Standard price	13	CA	APR-CA
32	A	Fitness Tracker	Garmin	£ 130.00	5	Electronics	Standard price	24	CA	MAY-CA
33	A	Jeans	Levi's	£ 50.00	50	Fashion	Standard price	02	CA	DEC-CA
34	N	Laptop	HP	£ 950.00	25	Electronics	High Price	17	IN	JUN-IN
35	R	Watch	Casio	£ 100.00	20	Accessories	Standard price	09	FR	JUL-FR

Step 4: Conditional Functions

- **SUMIF**
- **COUNTIF**

Total price of products in the Electronics category - (SUMIF).

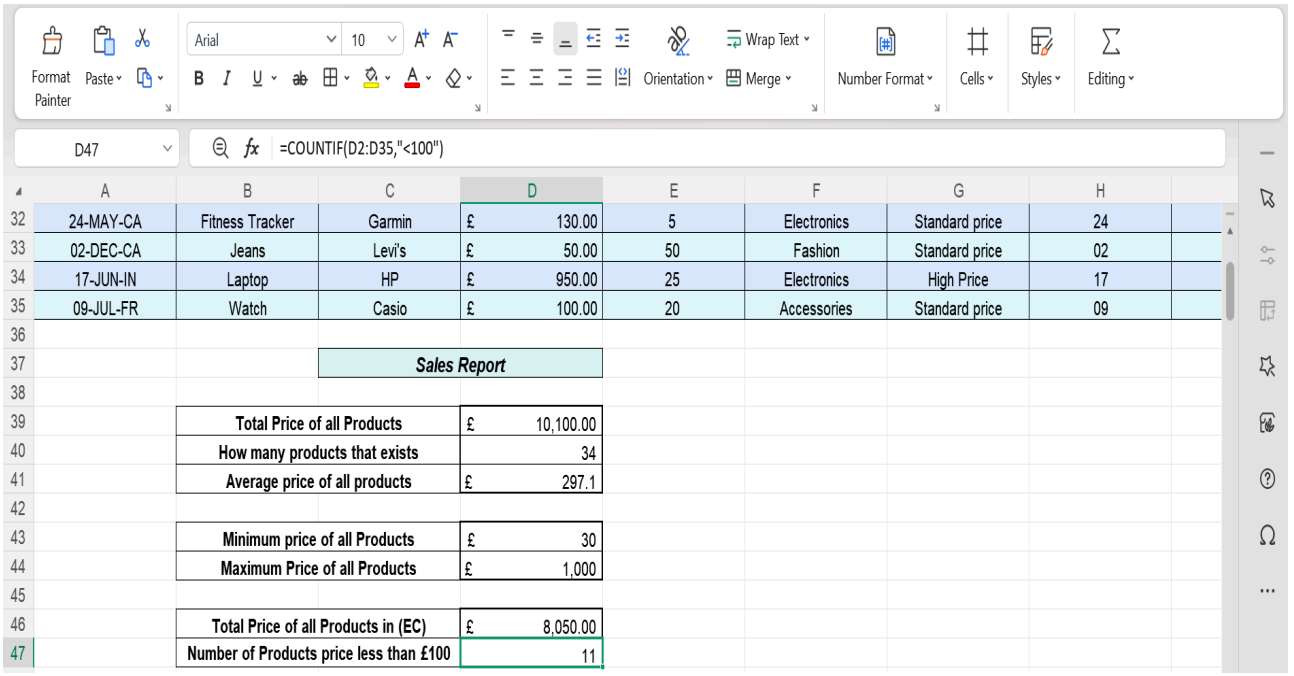


Format Painter | Paste | Arial | 10 | Wrap Text | Orientation | Merge | Number Format | Cells | Styles | Editing

D46 | =SUMIF(F2:F35,"Electronics",D2:D35)

	A	B	C	D	E	F	G	H
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	Electronics	Standard price	24
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	Fashion	Standard price	02
34	17-JUN-IN	Laptop	HP	£ 950.00	25	Electronics	High Price	17
35	09-JUL-FR	Watch	Casio	£ 100.00	20	Accessories	Standard price	09
36								
37			***Sales Report***					
38								
39		**Total Price of all Products**		£ 10,100.00				
40		**How many products that exists**		34				
41		**Average price of all products**		£ 297.1				
42								
43		**Minimum price of all Products**		£ 30				
44		**Maximum Price of all Products**		£ 1,000				
45								
46		**Total Price of all Products in (EC)**		£ 8,050.00				
47		**Number of Products price less than £100**		11				

Number of products with a price less than $100 - (COUNTIF)



Format Painter | Paste | Arial | 10 | Wrap Text | Orientation | Merge | Number Format | Cells | Styles | Editing

D47 | =COUNTIF(D2:D35,"<100")

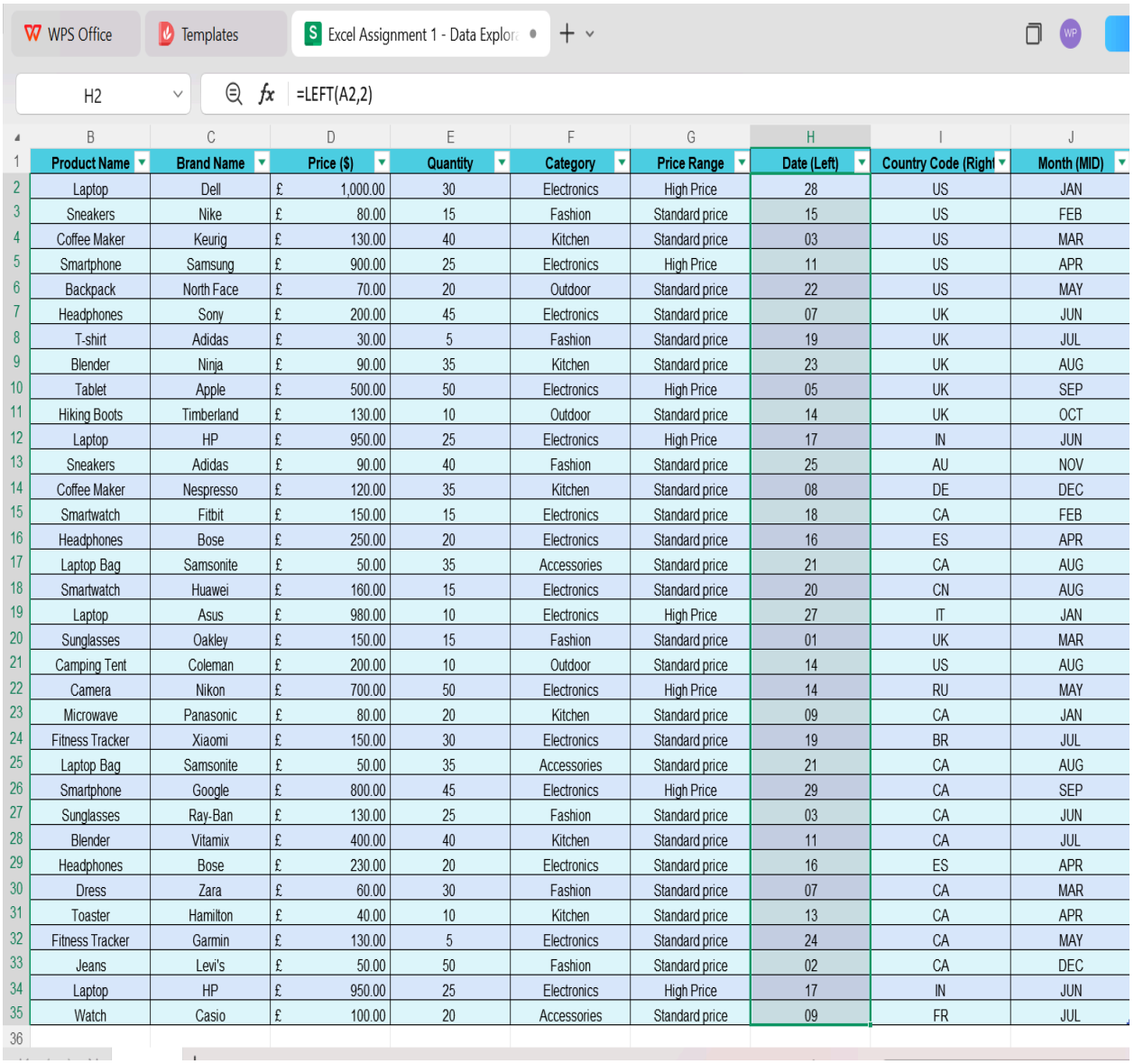
	A	B	C	D	E	F	G	H
32	24-MAY-CA	Fitness Tracker	Garmin	£ 130.00	5	Electronics	Standard price	24
33	02-DEC-CA	Jeans	Levi's	£ 50.00	50	Fashion	Standard price	02
34	17-JUN-IN	Laptop	HP	£ 950.00	25	Electronics	High Price	17
35	09-JUL-FR	Watch	Casio	£ 100.00	20	Accessories	Standard price	09
36								
37			***Sales Report***					
38								
39		**Total Price of all Products**		£ 10,100.00				
40		**How many products that exists**		34				
41		**Average price of all products**		£ 297.1				
42								
43		**Minimum price of all Products**		£ 30				
44		**Maximum Price of all Products**		£ 1,000				
45								
46		**Total Price of all Products in (EC)**		£ 8,050.00				
47		**Number of Products price less than £100**		11				

Step 5: Text Formatting

- **LEFT FUNCTION**
- **RIGHT FUNCTION**
- **MIDDLE FUNCTION**

'Product ID' using the (LEFT) function.

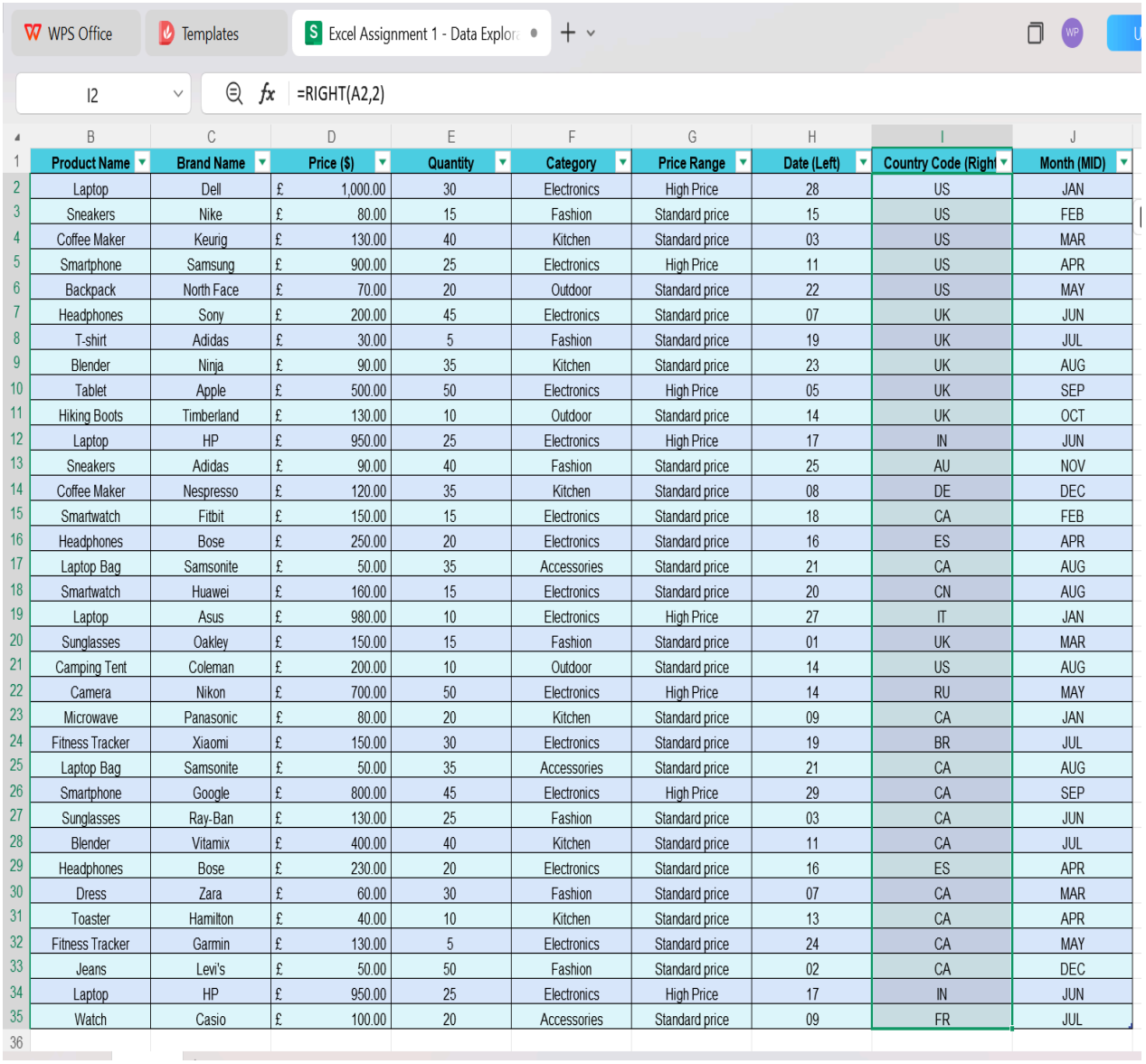
Left Function:



	B	C	D	E	F	G	H	I	J
1	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range	Date (Left)	Country Code (Right	Month (MID)
2	Laptop	Dell	£ 1,000.00	30	Electronics	High Price	28	US	JAN
3	Sneakers	Nike	£ 80.00	15	Fashion	Standard price	15	US	FEB
4	Coffee Maker	Keurig	£ 130.00	40	Kitchen	Standard price	03	US	MAR
5	Smartphone	Samsung	£ 900.00	25	Electronics	High Price	11	US	APR
6	Backpack	North Face	£ 70.00	20	Outdoor	Standard price	22	US	MAY
7	Headphones	Sony	£ 200.00	45	Electronics	Standard price	07	UK	JUN
8	T-shirt	Adidas	£ 30.00	5	Fashion	Standard price	19	UK	JUL
9	Blender	Ninja	£ 90.00	35	Kitchen	Standard price	23	UK	AUG
10	Tablet	Apple	£ 500.00	50	Electronics	High Price	05	UK	SEP
11	Hiking Boots	Timberland	£ 130.00	10	Outdoor	Standard price	14	UK	OCT
12	Laptop	HP	£ 950.00	25	Electronics	High Price	17	IN	JUN
13	Sneakers	Adidas	£ 90.00	40	Fashion	Standard price	25	AU	NOV
14	Coffee Maker	Nespresso	£ 120.00	35	Kitchen	Standard price	08	DE	DEC
15	Smartwatch	Fitbit	£ 150.00	15	Electronics	Standard price	18	CA	FEB
16	Headphones	Bose	£ 250.00	20	Electronics	Standard price	16	ES	APR
17	Laptop Bag	Samsonite	£ 50.00	35	Accessories	Standard price	21	CA	AUG
18	Smartwatch	Huawei	£ 160.00	15	Electronics	Standard price	20	CN	AUG
19	Laptop	Asus	£ 980.00	10	Electronics	High Price	27	IT	JAN
20	Sunglasses	Oakley	£ 150.00	15	Fashion	Standard price	01	UK	MAR
21	Camping Tent	Coleman	£ 200.00	10	Outdoor	Standard price	14	US	AUG
22	Camera	Nikon	£ 700.00	50	Electronics	High Price	14	RU	MAY
23	Microwave	Panasonic	£ 80.00	20	Kitchen	Standard price	09	CA	JAN
24	Fitness Tracker	Xiaomi	£ 150.00	30	Electronics	Standard price	19	BR	JUL
25	Laptop Bag	Samsonite	£ 50.00	35	Accessories	Standard price	21	CA	AUG
26	Smartphone	Google	£ 800.00	45	Electronics	High Price	29	CA	SEP
27	Sunglasses	Ray-Ban	£ 130.00	25	Fashion	Standard price	03	CA	JUN
28	Blender	Vitamix	£ 400.00	40	Kitchen	Standard price	11	CA	JUL
29	Headphones	Bose	£ 230.00	20	Electronics	Standard price	16	ES	APR
30	Dress	Zara	£ 60.00	30	Fashion	Standard price	07	CA	MAR
31	Toaster	Hamilton	£ 40.00	10	Kitchen	Standard price	13	CA	APR
32	Fitness Tracker	Garmin	£ 130.00	5	Electronics	Standard price	24	CA	MAY
33	Jeans	Levi's	£ 50.00	50	Fashion	Standard price	02	CA	DEC
34	Laptop	HP	£ 950.00	25	Electronics	High Price	17	IN	JUN
35	Watch	Casio	£ 100.00	20	Accessories	Standard price	09	FR	JUL
36									

The 'Product ID' column using the (RIGHT) function.

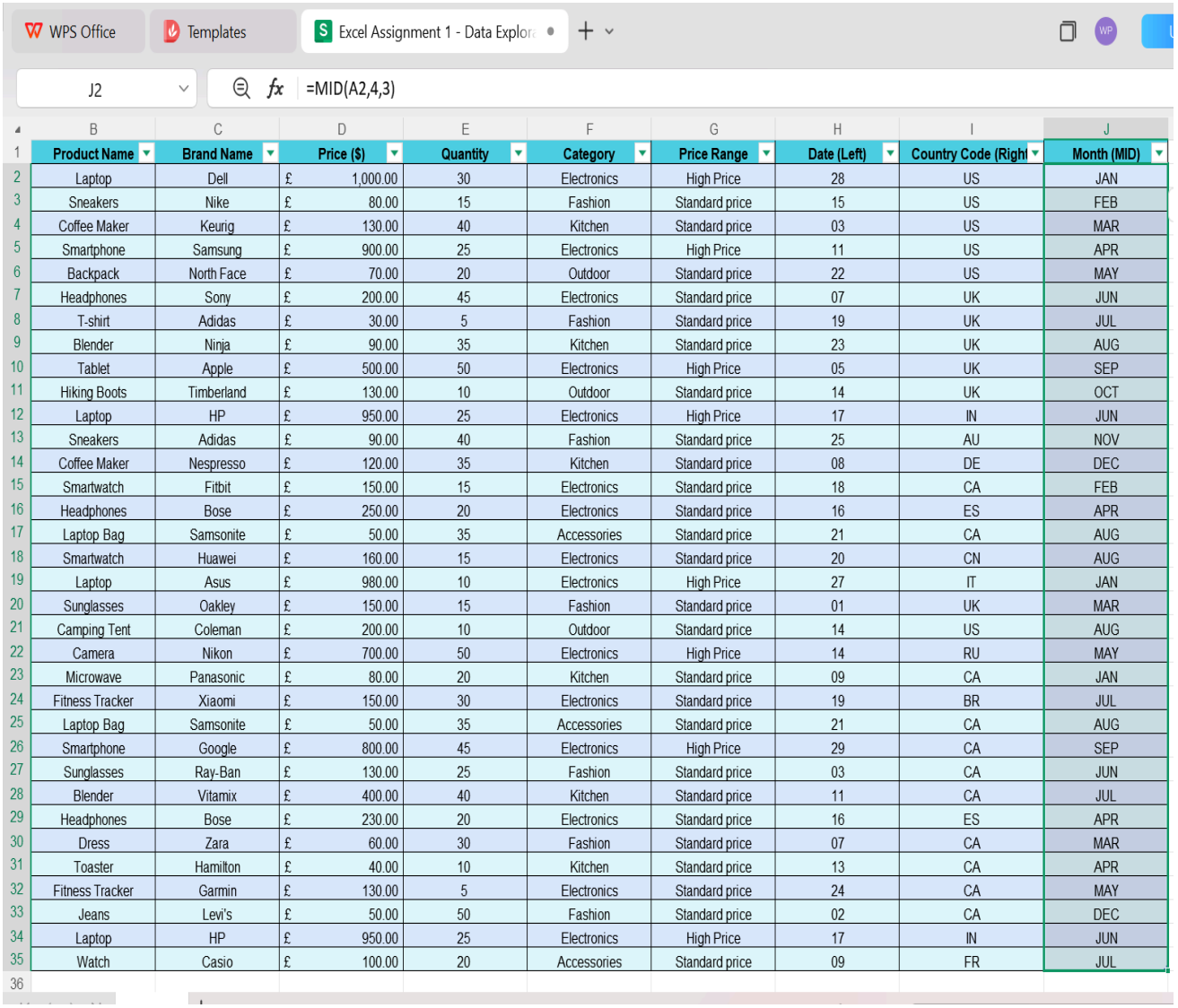
Right Function:



	B	C	D	E	F	G	H	I	J
1	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range	Date (Left)	Country Code (Right	Month (MID)
2	Laptop	Dell	£ 1,000.00	30	Electronics	High Price	28	US	JAN
3	Sneakers	Nike	£ 80.00	15	Fashion	Standard price	15	US	FEB
4	Coffee Maker	Keurig	£ 130.00	40	Kitchen	Standard price	03	US	MAR
5	Smartphone	Samsung	£ 900.00	25	Electronics	High Price	11	US	APR
6	Backpack	North Face	£ 70.00	20	Outdoor	Standard price	22	US	MAY
7	Headphones	Sony	£ 200.00	45	Electronics	Standard price	07	UK	JUN
8	T-shirt	Adidas	£ 30.00	5	Fashion	Standard price	19	UK	JUL
9	Blender	Ninja	£ 90.00	35	Kitchen	Standard price	23	UK	AUG
10	Tablet	Apple	£ 500.00	50	Electronics	High Price	05	UK	SEP
11	Hiking Boots	Timberland	£ 130.00	10	Outdoor	Standard price	14	UK	OCT
12	Laptop	HP	£ 950.00	25	Electronics	High Price	17	IN	JUN
13	Sneakers	Adidas	£ 90.00	40	Fashion	Standard price	25	AU	NOV
14	Coffee Maker	Nespresso	£ 120.00	35	Kitchen	Standard price	08	DE	DEC
15	Smartwatch	Fitbit	£ 150.00	15	Electronics	Standard price	18	CA	FEB
16	Headphones	Bose	£ 250.00	20	Electronics	Standard price	16	ES	APR
17	Laptop Bag	Samsonite	£ 50.00	35	Accessories	Standard price	21	CA	AUG
18	Smartwatch	Huawei	£ 160.00	15	Electronics	Standard price	20	CN	AUG
19	Laptop	Asus	£ 980.00	10	Electronics	High Price	27	IT	JAN
20	Sunglasses	Oakley	£ 150.00	15	Fashion	Standard price	01	UK	MAR
21	Camping Tent	Coleman	£ 200.00	10	Outdoor	Standard price	14	US	AUG
22	Camera	Nikon	£ 700.00	50	Electronics	High Price	14	RU	MAY
23	Microwave	Panasonic	£ 80.00	20	Kitchen	Standard price	09	CA	JAN
24	Fitness Tracker	Xiaomi	£ 150.00	30	Electronics	Standard price	19	BR	JUL
25	Laptop Bag	Samsonite	£ 50.00	35	Accessories	Standard price	21	CA	AUG
26	Smartphone	Google	£ 800.00	45	Electronics	High Price	29	CA	SEP
27	Sunglasses	Ray-Ban	£ 130.00	25	Fashion	Standard price	03	CA	JUN
28	Blender	Vitamix	£ 400.00	40	Kitchen	Standard price	11	CA	JUL
29	Headphones	Bose	£ 230.00	20	Electronics	Standard price	16	ES	APR
30	Dress	Zara	£ 60.00	30	Fashion	Standard price	07	CA	MAR
31	Toaster	Hamilton	£ 40.00	10	Kitchen	Standard price	13	CA	APR
32	Fitness Tracker	Garmin	£ 130.00	5	Electronics	Standard price	24	CA	MAY
33	Jeans	Levi's	£ 50.00	50	Fashion	Standard price	02	CA	DEC
34	Laptop	HP	£ 950.00	25	Electronics	High Price	17	IN	JUN
35	Watch	Casio	£ 100.00	20	Accessories	Standard price	09	FR	JUL
36									

The 'Product ID' column using the(MID) function.

Middle Function:



	B	C	D	E	F	G	H	I	J
1	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range	Date (Left)	Country Code (Right	Month (MID)
2	Laptop	Dell	£ 1,000.00	30	Electronics	High Price	28	US	JAN
3	Sneakers	Nike	£ 80.00	15	Fashion	Standard price	15	US	FEB
4	Coffee Maker	Keurig	£ 130.00	40	Kitchen	Standard price	03	US	MAR
5	Smartphone	Samsung	£ 900.00	25	Electronics	High Price	11	US	APR
6	Backpack	North Face	£ 70.00	20	Outdoor	Standard price	22	US	MAY
7	Headphones	Sony	£ 200.00	45	Electronics	Standard price	07	UK	JUN
8	T-shirt	Adidas	£ 30.00	5	Fashion	Standard price	19	UK	JUL
9	Blender	Ninja	£ 90.00	35	Kitchen	Standard price	23	UK	AUG
10	Tablet	Apple	£ 500.00	50	Electronics	High Price	05	UK	SEP
11	Hiking Boots	Timberland	£ 130.00	10	Outdoor	Standard price	14	UK	OCT
12	Laptop	HP	£ 950.00	25	Electronics	High Price	17	IN	JUN
13	Sneakers	Adidas	£ 90.00	40	Fashion	Standard price	25	AU	NOV
14	Coffee Maker	Nespresso	£ 120.00	35	Kitchen	Standard price	08	DE	DEC
15	Smartwatch	Fitbit	£ 150.00	15	Electronics	Standard price	18	CA	FEB
16	Headphones	Bose	£ 250.00	20	Electronics	Standard price	16	ES	APR
17	Laptop Bag	Samsonite	£ 50.00	35	Accessories	Standard price	21	CA	AUG
18	Smartwatch	Huawei	£ 160.00	15	Electronics	Standard price	20	CN	AUG
19	Laptop	Asus	£ 980.00	10	Electronics	High Price	27	IT	JAN
20	Sunglasses	Oakley	£ 150.00	15	Fashion	Standard price	01	UK	MAR
21	Camping Tent	Coleman	£ 200.00	10	Outdoor	Standard price	14	US	AUG
22	Camera	Nikon	£ 700.00	50	Electronics	High Price	14	RU	MAY
23	Microwave	Panasonic	£ 80.00	20	Kitchen	Standard price	09	CA	JAN
24	Fitness Tracker	Xiaomi	£ 150.00	30	Electronics	Standard price	19	BR	JUL
25	Laptop Bag	Samsonite	£ 50.00	35	Accessories	Standard price	21	CA	AUG
26	Smartphone	Google	£ 800.00	45	Electronics	High Price	29	CA	SEP
27	Sunglasses	Ray-Ban	£ 130.00	25	Fashion	Standard price	03	CA	JUN
28	Blender	Vitamix	£ 400.00	40	Kitchen	Standard price	11	CA	JUL
29	Headphones	Bose	£ 230.00	20	Electronics	Standard price	16	ES	APR
30	Dress	Zara	£ 60.00	30	Fashion	Standard price	07	CA	MAR
31	Toaster	Hamilton	£ 40.00	10	Kitchen	Standard price	13	CA	APR
32	Fitness Tracker	Garmin	£ 130.00	5	Electronics	Standard price	24	CA	MAY
33	Jeans	Levi's	£ 50.00	50	Fashion	Standard price	02	CA	DEC
34	Laptop	HP	£ 950.00	25	Electronics	High Price	17	IN	JUN
35	Watch	Casio	£ 100.00	20	Accessories	Standard price	09	FR	JUL
36									